

AI Cluster Benchmarks - Establishing a standardized evaluation system for Superpods

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Standard Representative Huawei



The Need for Next-Generation Computing System Benchmarks

Changes in the hardware form of computing devices

Changes in workloads and system architecture

Changes in the computing system software

New Clusters – Balancing Current Demands and Future Superpod Evolution



CPU-Centric Primary-Secondary Architecture → Intelligent Computing Superpod

CPU-Centric Primary-Secondary Architecture → General-Purpose Computing Superpod

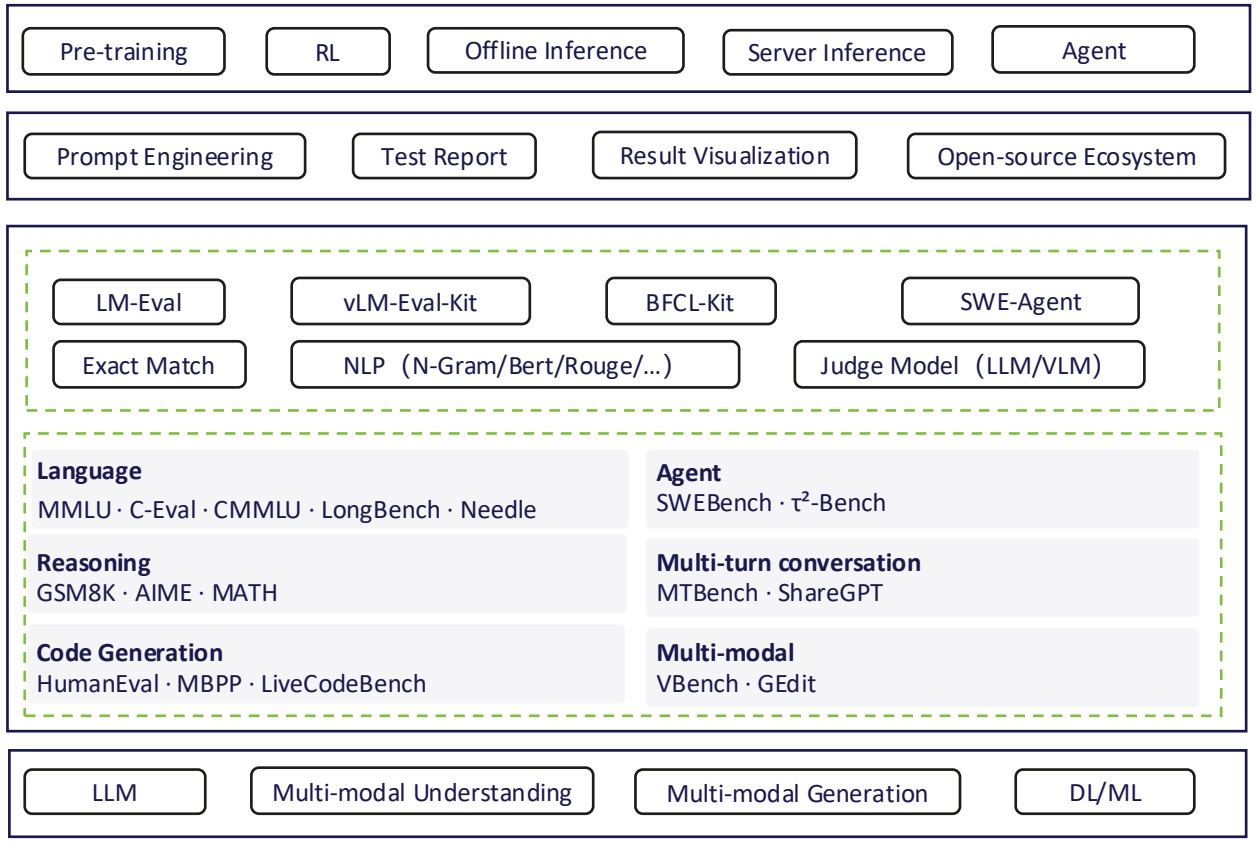
Why Two Benchmarks?

AISBENCH
Focus: AI Training & Inference
Goal: Comprehensive Metrics & Industry-Oriented Workloads to Uncover System Bottlenecks

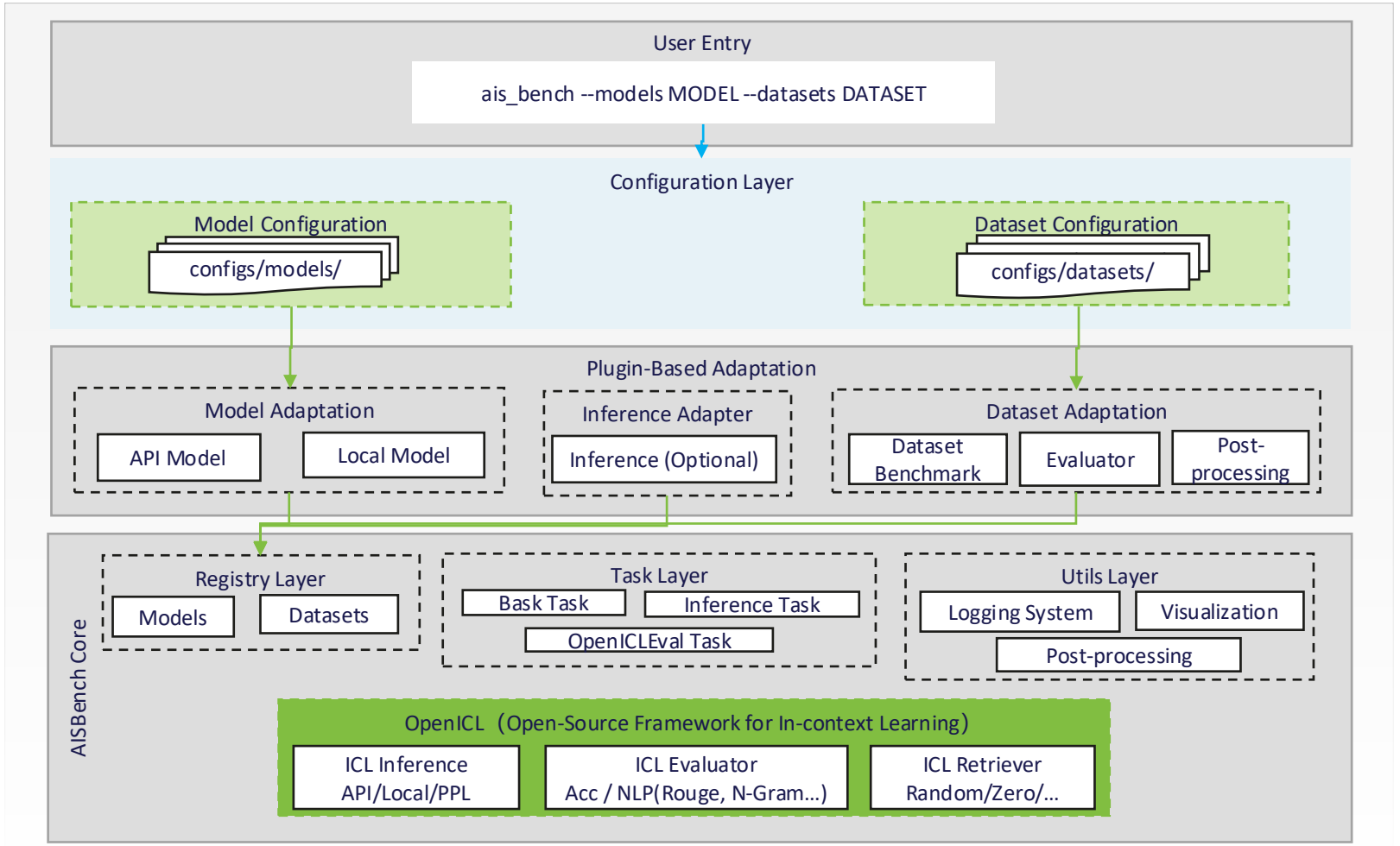
CLUSTERBENCH
Focus: Big Data & Cloud
Goal: Interconnect-Driven Full-Stack Capability & Effective Computing Capacity

AISBench: AI System Performance Benchmark

- A comprehensive evaluation framework for AI computing centers – covering performance, reliability, compute efficiency, and energy efficiency
- OpenICL-based decoupling of dataset loading, inference, and evaluation – fast adaptation to new workloads
- Full coverage of mainstream models and 70+ datasets



AISBench Architecture



② **Launch evaluation scripts**

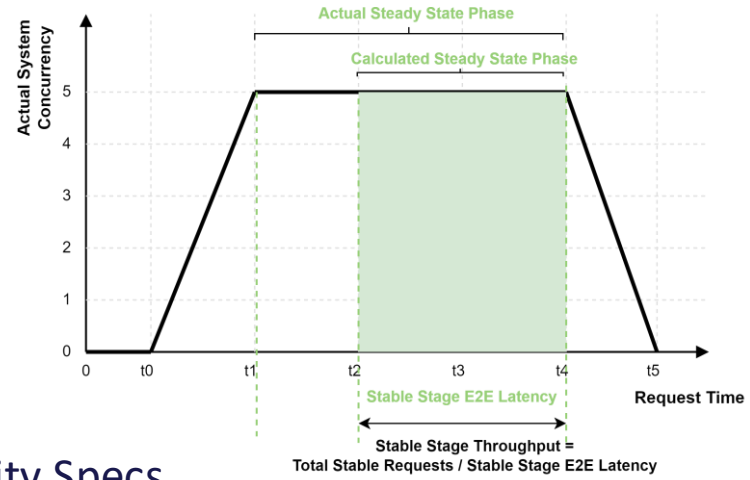
① **Integrate new models / new datasets**

③ **Execute evaluation tasks concurrently**

Three-Tier Cluster Benchmarking Framework

- L2: Application-level metrics
  - Training/inference throughput and duration
  - Compute efficiency per FLOP
  - Cluster utilization
  - Cluster availability/reliability
- L1: System-level technical metrics
  - Collective communication bandwidth
  - Chip compute efficiency
  - Cluster linearity
  - Bandwidth utilization (communication & memory)
  - MTTR
- L0: Hardware capabilities
  - Cluster computing power
  - Cluster memory access
  - Cluster and datacenter interconnect
  - Cluster storage
  - resource pooling

High-Concurrency Stable-Stage Performance Evaluation



Under identical request load (fixed concurrency, varying I/O lengths), stable-stage QPM shows a 6% improvement over the entire phase average.

Full-Stage Statistics (QPM: 397)

Common Metric	Stage	Value
Benchmark Duration	total	773599.3664 ms
Total Requests	total	20480
Failed Requests	total	0
Success Requests	total	20480
Concurrency	total	1399.5763
Max Concurrency	total	1500
Request Throughput	total	26.4737 req/s

Stable-Stage Statistics (QPM: 420)

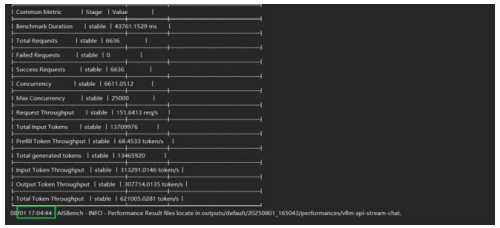
Benchmark Duration	stable	649219.4909 ms
Total Requests	stable	18213
Failed Requests	stable	0
Success Requests	stable	18213
Concurrency	stable	1491.7118
Max Concurrency	stable	1500
Request Throughput	stable	28.0537 req/s

Capacity Specs

Total Requests	Input Tokens per Request	Output Tokens per Request	Max Memory Usage (GB)	Max Disk Usage (GB)	Data Compute Time (s)
10000	1024	1024	< 16	0.12	3
10000	1024	4096	< 16	0.16	4
10000	4096	4096	< 16	0.17	6
50000	4096	4096	< 32	0.80	30
250000	4096	4096	< 64	4.00	150

High-Concurrency Inference – Supports up to 250K requests in superpod scenarios, with minute-level metric computation.





Code Agent Benchmarking

- A test to check whether superpods can fulfill the promise of providing cost-effective inference computing power for the agent era

Multi-turn Interaction

Long Context

High Concurrency

τ^2-Bench

Evaluating conversational AI agents in real-world, tool-driven, **dual-control** scenarios where both the agent and a simulated user coordinate actions to complete complex customer service tasks

Airline **Retail** **Telecom**

5 Checks

db_check | tool call | policy | goal | dialogue

Pass[1] Pass[k]

SWE-bench

Evaluating the performance of LLMs on **real-world software issues**: Given a codebase and an issue, generate a patch to solve the problem

5 Datasets

full | verified | verified_mini | lite | multilingual

2 Tasks

infer Calls mini-swe-agent to generate a patch
eval Calls SWE-bench harness to collect statistics on resolved

accuracy
resolved / total

--reuse
Breakpoint resume

Harbor

Evaluating the comprehensive capabilities of task completion, tool usage, and policy compliance

Multiple Agents

terminus-2 | claude-code | openhands | aider | codex | Custom agent

Multiple Environments

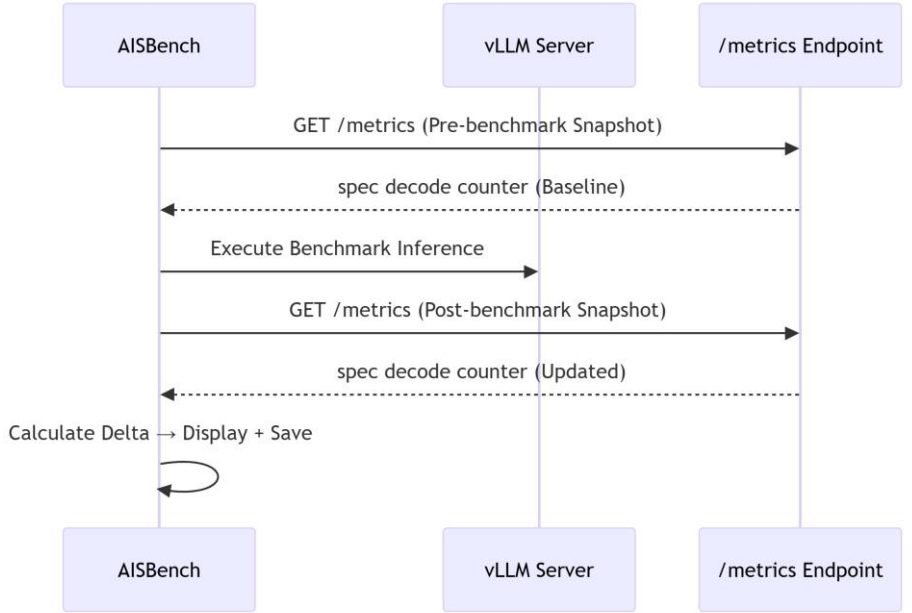
Docker | Daytona | E2B | Modal

Avg Score pass@k

Code Agent: Speculative Decoding

- When evaluating the performance (--mode perf) of a vLLM-compatible inference service with speculative decoding enabled, AISBench can collect the speculative decoding performance counters of the server through the Prometheus /metrics endpoint. This allows for the collection of decoding efficiency metrics in addition to the standard request-level latency and throughput metrics.

Metric	Description
Drafts	Number of draft-verify cycles within the evaluation window
Draft tokens	Total number of candidate tokens generated by the draft model
Accepted tokens	Total number of tokens accepted by the target model
Acceptance rate (%)	(Accepted tokens / Draft tokens) × 100
Acceptance length	1 + (Accepted tokens / Number of drafts) — average number of accepted tokens per forward pass
Per-position rates	Acceptance rate at each draft position



ClusterBench: Performance Evaluation for General-Computing Superpods

- **ClusterBench:** A performance evaluation tool for rack-scale and cluster systems. Focused on general-computing superpods, pooled architectures, and multi-node systems to deliver systematic and highly accurate cluster benchmarking.
- **Target Sectors:** Cloud & Enterprise, Telecom, Hardware Ecosystem
- **Use Cases：** Pre-sales POC、 Deployment Optimization、 Capacity Planning、 Bottleneck Analysis

SUT

Integrated Performance Evaluation：

- **Server：** CPU、Memory、Storage、NIC、DPU…
- **OS：** Compilers、Schedulers、Runtimes、Accelerated Libraries…

Test Nodes: 1 to N Nodes

Interconnect Architectures： CXL-Based Systems/UB-Based Systems/Ethernet-Extended Systems （arm+x86+……）

Application Scenarios

Core Capabilities

Multi-Dimensional Evaluation Model

Overall Representation：

- Rack/Cluster Overall Performance (Effective Compute Capacity)
- Application Performance: Business performance across application scenarios

Detailed Sub-dimensions：

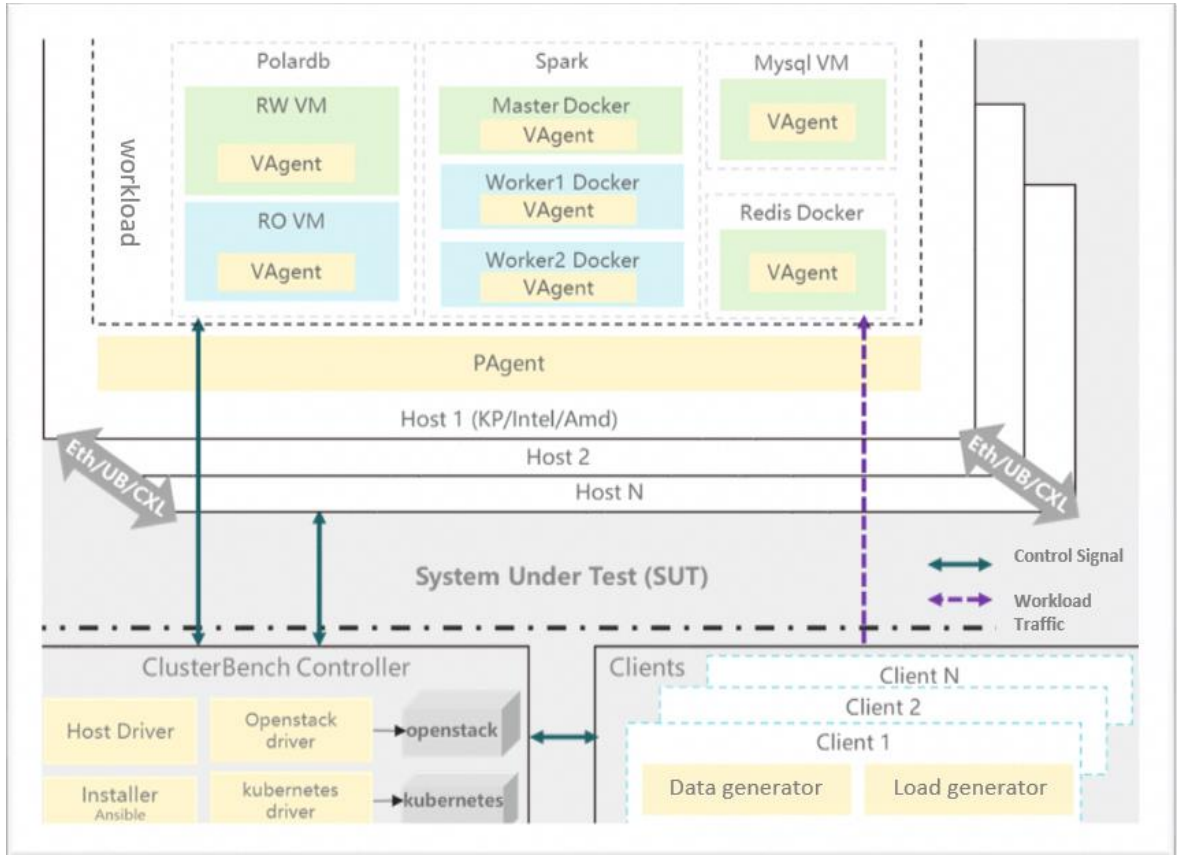
- **Application / Cluster Scalability**
- **Quality of Service (QoS) / Reliability:** P90 response time
- **Computing Capacity:** integer, floating-point
- **Memory Access Capacity:** bandwidth, latency, I/O capacity, etc.

ClusterBench Technical Overview

Performance Metrics & Evaluation	Cost-Performance Metric Definition	L2 Service Performance Metric Suite	L1 System Characteristic Metric Suite	L0 Baseline Specification Metric Suite	Metric Analytics & Presentation Precision/Repeatability
Workload Selection & Analytics	Workload Selection Representativeness/Versatility	Workload Feature Analysis Feature Diversity/Modeling	Performance Data Analytics	Workload Modeling Networking/Traffic/Stress Model/Datasets	Workload DFX Compatibility/Stability/Fairness
Traffic Patterns & Execution Engine	Trough Traffic Testing	Burst Traffic Testing	Peak Traffic Testing	Workload Stress Testing Engine	Workload Hybrid Deployment
Cloud & Virtualization Management	Virtualization Platform Adaptation openstack	Container Platform Adaptation k8s	Live Migration Control	Overcommitment Control	Virtual Node Scheduling
Automation & Orchestration	Automated Installation & Deployment	Node Communication & Management	Test Workflow Control	Benchmarking Reports	Acceleration Library Enablement
Infrastructure & Hardware Ecosystem	OS Adaptation openeuler/kylinOS/UOS	Hardware Architecture Adaptation X86/Aarch64			Interconnect Topology Adaptation CXL/ UB/TCP

Testing Framework

- Supports one-click automated deployment and benchmarking, compatible with VM and container workload deployments

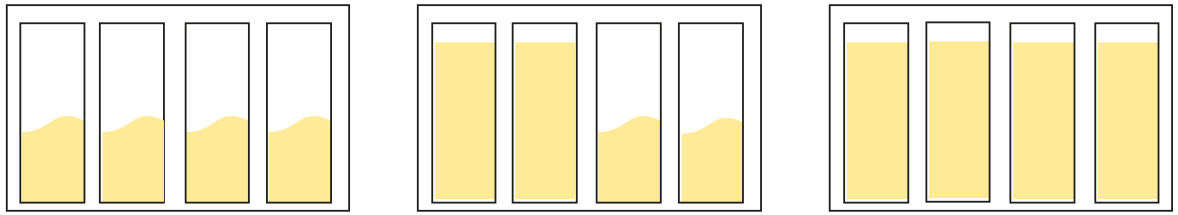


Cluster Benchmarking Support	**Node Control** Pagent / VAgent	**Runtime Monitoring** Periodic Collection & Reporting
VM & Container Workload Deployment	**Cloud Management Control** K8S / OpenStack	**Overcommitment Control** Overcommit Ratio Configuration
One-Click Installation, Deployment & Execution	**Compatibility** Unified Workflow	**Automated Deployment** Multi-Node Concurrent Deployment
Flexible Benchmarking Methodologies	**Sysbench-like Mode**	**SPECvirt-like Mode**

Test Mode: Three-phase simulation

Three-phase simulation of peak/trough business load variations across different time windows to closely align with real-world Data Center (DC) characteristics

Stress Testing Flow



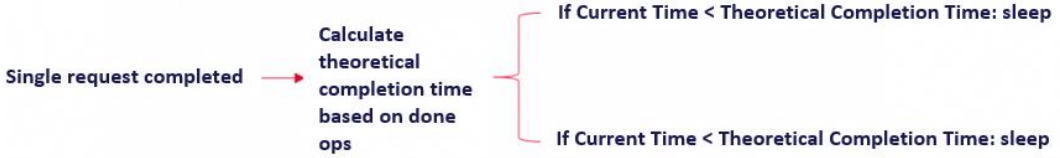
	Trough Phase:	Burst Phase:	Peak Phase:
Performance Evaluation Metrics	**Request Latency**	**High-load instances:** Request Throughput **Low-load instances:** Request Latency	**QoS-compliant Request Throughput, Resource Fragmentation (Memory, CPU)**
Cluster Design Directives	**Lower Latency**	**Resource Utilization, Load Balancing**	**High Bandwidth & Capacity Concurrent Processing Capability Load Scheduling Capability**

2 Load Control Methods

- **Fixed Time Interval Between Consecutive Requests Within a Thread**



- **Fixed Request Rate Issued Per Second by Benchmark Client (ops)**



3 Burst Modes

- **2 configuration options: Burst Ratio, Burst Workload**
- **Mode 1 —** Tile-based: Set **Burst Ratio** to **X** (Burst Workload unset), Burst occurs in X% of Tiles.
- **Mode 2 —** Workload-based: Set **Burst Workload** to **Y** (Burst Ratio unset) ,Workload Y in all Tiles undergoes burst.
- **Mode 3 —** Hybrid: Set **Burst Ratio** to **X**, **Burst Workload** to **Y** ,Workload Y in X% of Tiles undergoes burst.

Workload Selection: 6 major solution domains around "Internet + Public Livelihood"

ID	Workload	Component	Metrics	Characteristics
1	**Transaction Processing**	MySQL / PostgreSQL	Throughput, Latency	Small block random I/O, Lock synchronization
2	**In-Memory Database**	Redis	Throughput, Latency	Single-threaded, Sensitive to interrupt/context switching
3	**AI Agent**	openclaw	Success Rate, Latency	High statefulness, Long-tail latency
4	**Web Services**	nginx+ES+ SpintBoot+Mysql	Throughput, Latency	High concurrency, Long link invocations
5	**Cloud-Native Database**	PolarDB	Throughput, Latency	Primary-secondary, Read/write splitting
6	**Stream Processing**	Flink+kafka	Throughput	Stateful computing, Event streaming, Concurrent SQL
7	**Hotel Reservation**	microservices	Throughput, Latency	Microservices, Load balancing, Containerization
8	**TPCDS**	Hive on Spark	Throughput, Latency	Multi-operator Shuffle, Complex Join
9	**VM Live Migration**	Vm_migrate	Max service interruption duration during migration	Virtualization, Cold/hot page identification, Iterative migration
10	**RPC Communication**	bRPC	Throughput, Latency	Latency-sensitive, Multi-hop forwarding

Three-Tier Cluster Benchmarking Framework

Database： TPC-C； Database RTO；

Virtualization： Resource utilization rate (CPU, memory bandwidth),RPC communication latency, High-spec VM & resource utilization rate, VM density;

Search & Recommendation: RPC communication latency; Memory bandwidth, read/write throughput;

Financial Trading: Network communication latency; Data synchronization rate;

Database Appliance (All-in-One): High bandwidth, low-latency communication; Eliminates I/O bottlenecks;

Tier	Key Metrics	Example
L2	Business Throughput	ops/rps/tps=1468
	Business Latency	280ms
	VM Density	12 VM / Host
	Resource Allocation Ratio	CPU：150% Memory：110%
	Resource Utilization Rate	50%
L1	VM Migration Latency	100ms
	RPC Communication Latency	0.21ms
	Cross-Node Memory Access	250ns
	Migration Efficiency under Specific Dirty Page Rate	2min
L0	Total Cluster Compute Capacity	205
	Total Intra-Cluster Memory Access Bandwidth	4.5GB/s
	Inter-Node Interconnect Latency	1000ns
	Non/Cacheable Accessible Physical Memory Capacity	500GB

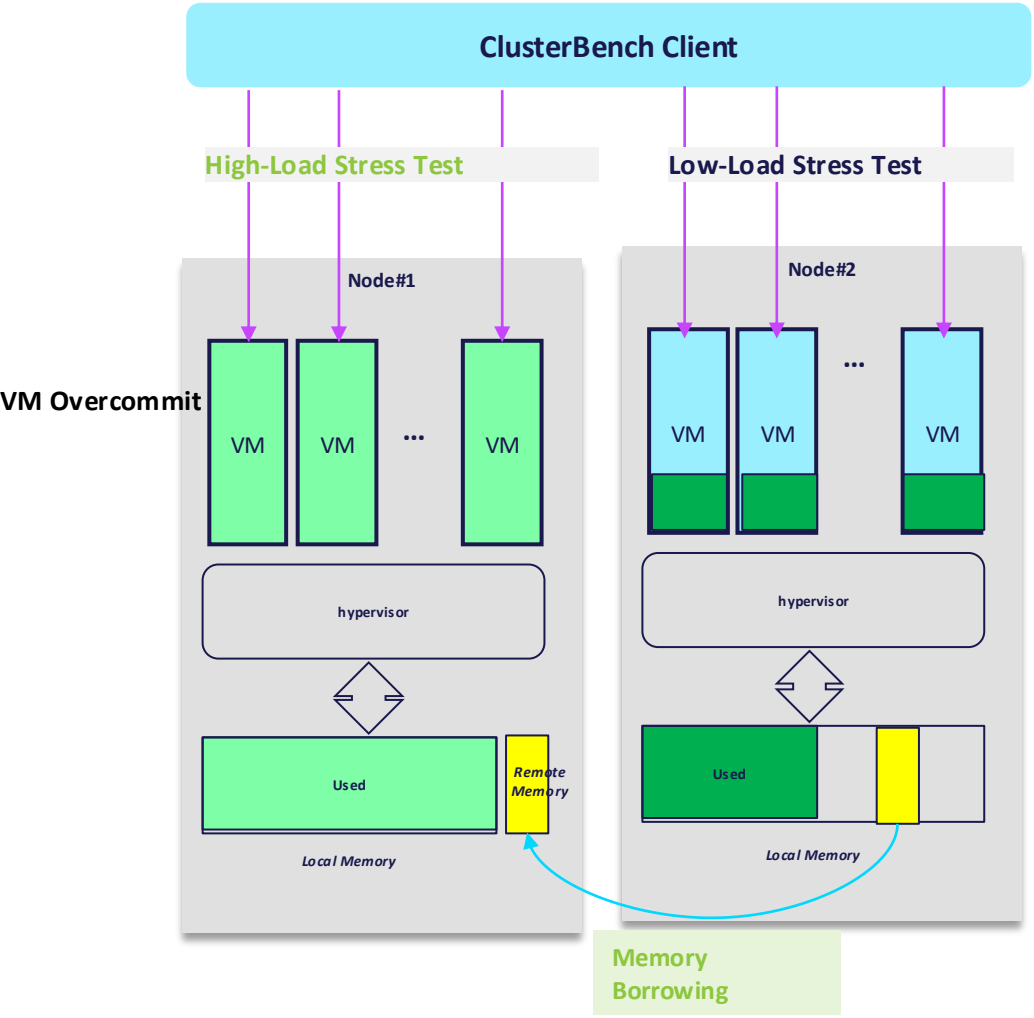
Test Environment Example

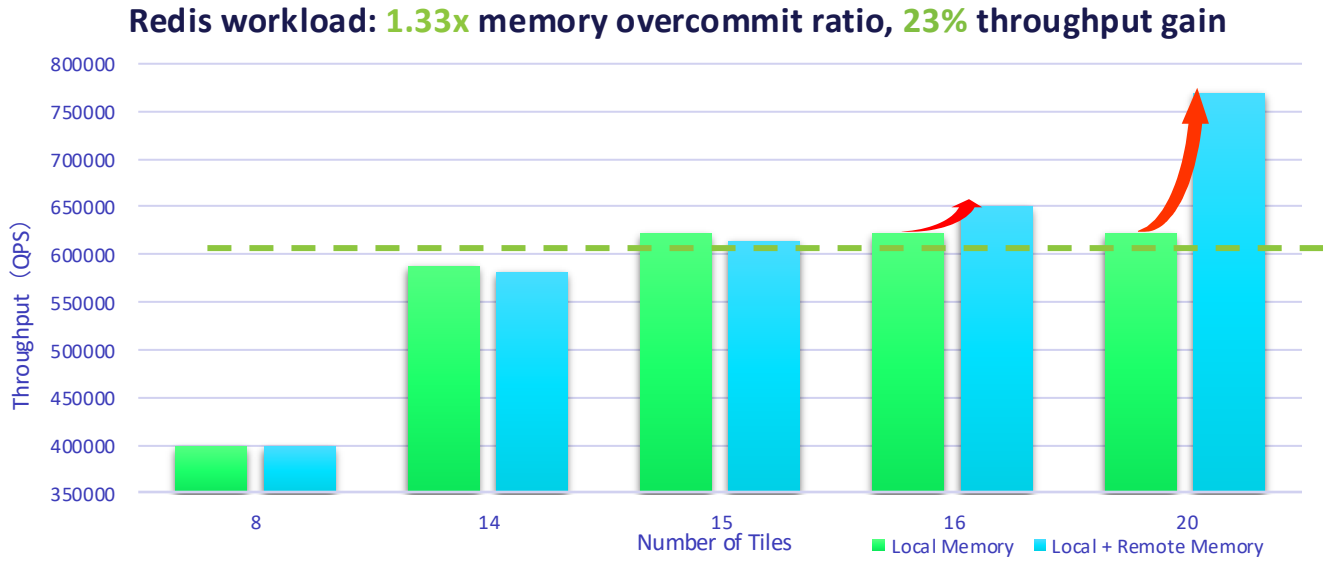
Environment	TCP	UB	CXL
Hardware	CPU：Kunpeng 920 Memory：24*96G 6400 MT/s Disk：nvme 7T Network：25GE NIC	CPU：Kunpeng 920 & UB Memory：24*96G 6400 MT/s Disk：nvme 7T	**Memory Pool:**： CXL Mxxxx CPU：intel atom cpu C3558 2.20GHz 4-Core Memory：4EP Ports * 4 Powertech ReTimer Cards, each EP port with 3 * 128GB Samsung DDR5 DIMMs **Server**： Mxxxx CPU：**AMD 9965 192core/ Intel Xeon 6780E** Memory：12*128GB Samsung DDR5 DIMMs Disk：nvme 1.7T Network：25GE NIC
OS / Kernel	openEuler 24.03 (LTS-SP3) Linux 6.6.0-142.0.0.127	openEuler 24.03 (LTS-SP3) Linux 6.6.0-142.0.0.127	**Memory Pool**： Centos Linux 7 (core) Linux H3G5 6.6.4 #282 **Server**： Ubuntu 24.04 LTS(Noble Numbat) Linux 6.8.0-31-generic #31-Ubuntu
Remote Memory Exposure Method	/	Presented as NUMA node	Presented as NUMA node
Component Version	/	BeiMing-I XXX	CXL 2.0

[Note]：

- **UB** is a unified interconnect protocol for SuperPods, aimed at integrating communications among I/O, memory access, and heterogeneous compute units (CPU/NPU/GPU, etc.) into a unified technical architecture to achieve a high-performance, highly collaborative, and elastic computing infrastructure.
- For more details about **UB**, refer to: https://www.unifiedbus.com/en

Case Study: Dynamically expanding node memory



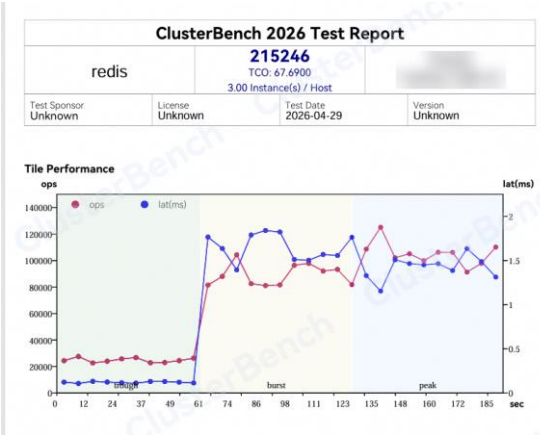


Openclaw workload: Supports up to 2.5 memory overcommit ratio

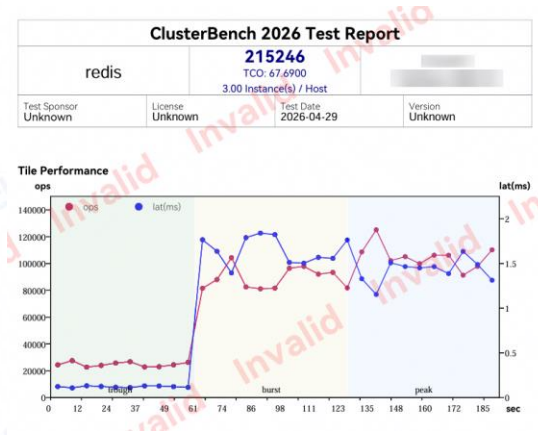
Openclaw Workload Evaluation Results					
VM Quantity (2U4G)	CPU Overcommit Ratio	Memory Overcommit Ratio	Task Success Rate	Avg Webpage Latency (s)	Borrowed Memory (GB)
100	2.08	1.25	100%	17.9	0
		1.5		18.3	40
		1.8		19.7	64
		2		22.9	72
		2.2		23.9	88
		2.5		28.3	101

Industry Support

ClusterBench 2026 Test Report			
redis	215246 TCO: 67.6900 3.00 Instance(s) / Host		
Test Sponsor Unknown	License	Test Date 2026-04-29	Version Unknown



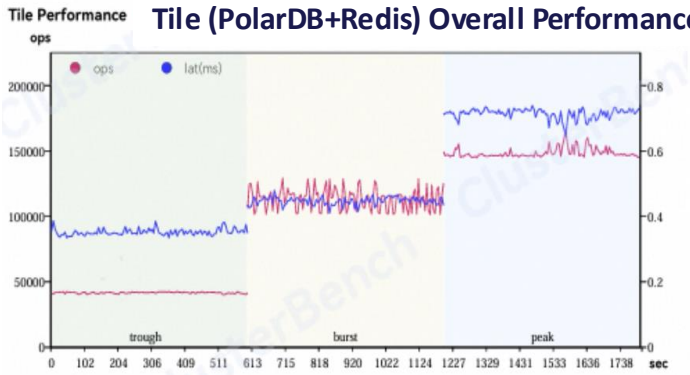
Valid Report



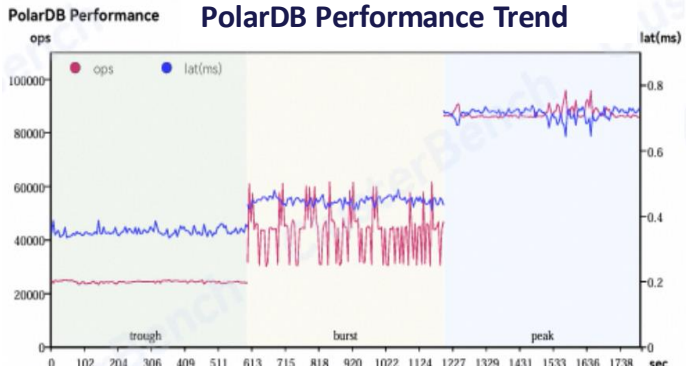
Invalid Report

- **Supports UB, CXL, and TCP**

Tile (PolarDB+Redis) Overall Performance Trend



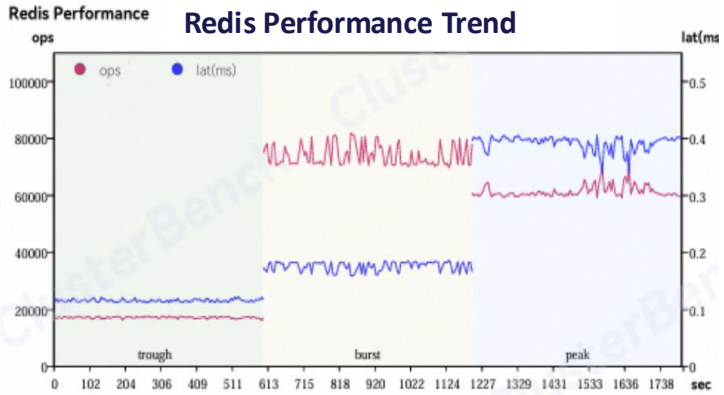
PolarDB Performance Trend



- **Released in GCC OAII community**

- **Supports 20+ vendors in POC**

Redis Performance Trend





Call to Action

- Build a More Open, Accurate, and Forward-Looking Evaluation Ecosystem
- Where to find additional information:
  - AISBench github: https://github.com/AISBench/benchmark
  - ClusterBench gitcode: https://gitcode.com/benchmark-sig/ClusterBench
  - Benchmark results will be uploaded here: https://openaiinfra.gccorg.com/projects/benchmark
- How to get involved in the project/sub-project community ：
  - contact baozhenzhong@gccorg.com, xuwenxi3@huawei.com

Industry-wide Collaboration

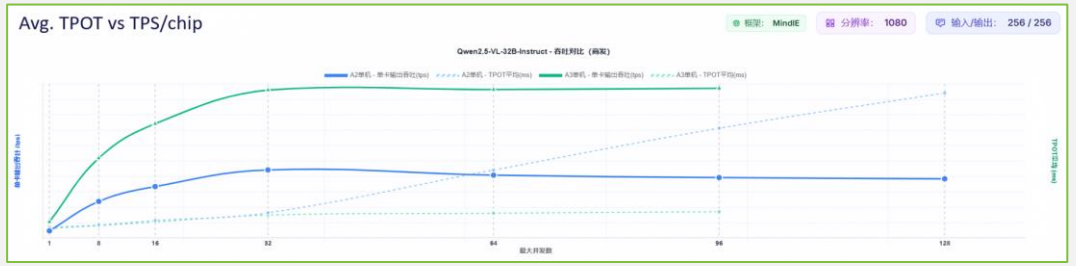
Community · Institute · 40+ customer adoptions

Inspur Intel Illuvatar Huawei CESI NITS GCC-OAII





Result Visualization



Thank You!

